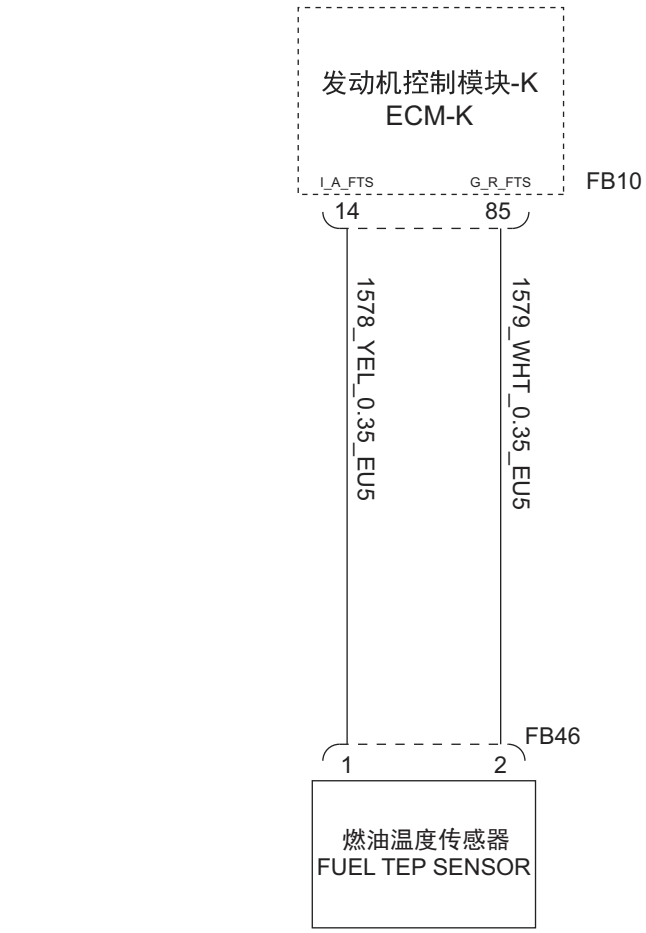
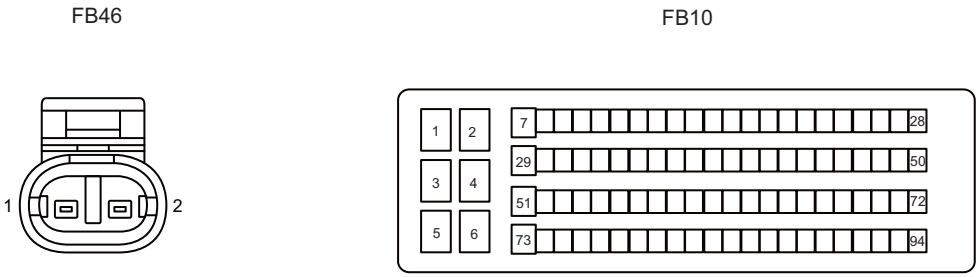
P0182、 P0183

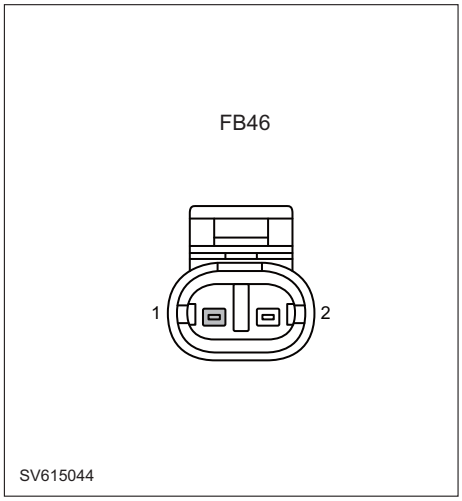
Description of fault diagnosis codes
P0182-11：SRC low for fuel temperature sensor
P0183-12：SRC high for fuel temperature sensor
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Fault in harness or harness plug • Fuel temperature sensor fault
Description of circuit diagram
The engine fuel temperature sensor is a variable resistor, which is used to measure the fuel temperature. The engine control module provides 5V voltage to the signal circuit of engine fuel temperature sensor and also provides grounding to low reference voltage circuit.

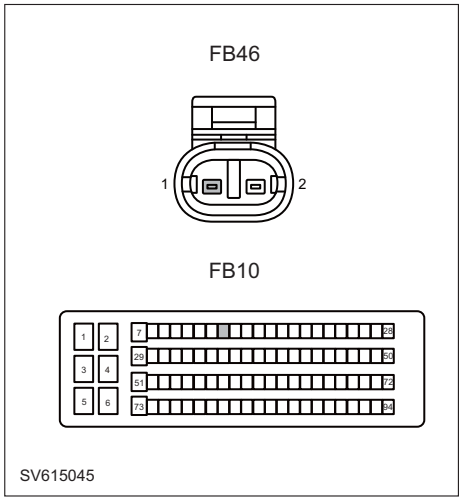
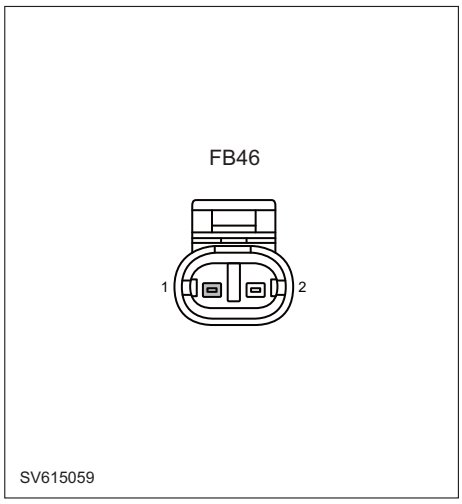
Circuit diagram

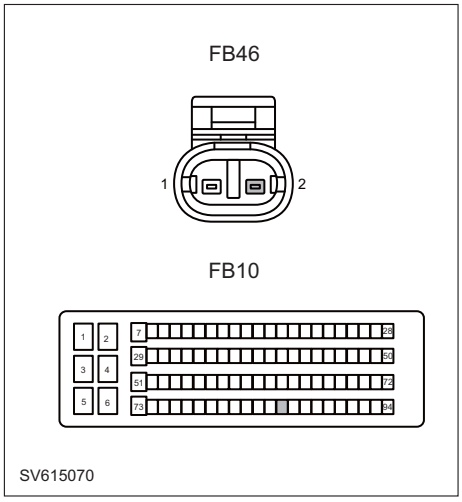


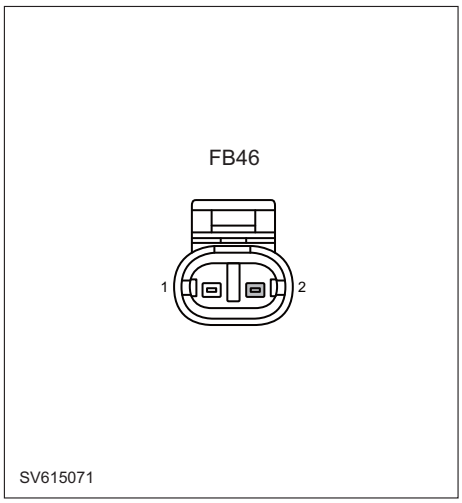


SV615027

Troubleshooting procedures		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the engine control module and the harness plug of accelerator pedal sensor for damage, poor contact, aging and looseness. 4 Check whether the result is normal.
		No → Repair or replace the faulted parts.
No ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to diagnosis interface. 2 Turn the ignition switch to ON. 3 Diagnose the engine control system with diagnostic apparatus. 4 Is there any other fault code except for P0182 and P0183?
		Yes → Reference:DTC Summary List (EU5).
No ↓		
Step 3	Check the signal circuit of fuel temperature sensor.	
		1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of fuel temperature sensor. 3 Turn the ignition switch to ON. 4 Measure the voltage between the harness plug of fuel temperature sensor and ground. Measuring circuit: the voltage between FB46 plug terminal 1 and ground. **Standard value:4.8~5.2V** 5 Check whether the result is normal?
		Yes → Go to Step 6.
No ↓		

Step 4	Check the signal circuit of fuel temperature sensor(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Measure the resistance of harness plug between fuel temperature sensor and engine control module. Measure circuit: the resistance between FB46 plug terminal 1 and FB10 plug terminal 14. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	
Step 5	Check the signal circuit of fuel temperature sensor(short circuit).
	1 Measure the resistance between the harness plug of fuel temperature sensor and ground. Measure circuit: the resistance between FB46 plug terminal 1 and ground. **Standard value: ∞** 2 Measure the voltage between the harness plug of fuel temperature sensor and ground. Measure circuit: the voltage between FB46 plug terminal 1 and ground. **Standard value: ≈ 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 6	Check the ground circuit of fuel temperature sensor(open circuit).
	1 Turn the ignition switch to OFF. 2 Disconnect the harness plug of engine control module. 3 Measure the resistance of harness plug between fuel temperature sensor and engine control module. Measure circuit: the resistance between FB46 plug terminal 2 and FB10 plug terminal 85. **Standard value: ≈ 0Ω** 4 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 7	Check the ground circuit of fuel temperature sensor(short circuit).
	1 Measure the resistance between the harness plug of fuel temperature sensor and ground. Measure circuit: the resistance between FB46 plug terminal 2 and ground. **Standard value: ∞** 2 Measure the voltage between the harness plug of fuel temperature sensor and ground. Measure circuit: the voltage between FB46 plug terminal 2 and ground. **Standard value: ≈ 0V** 3 Check whether the result is normal?
	No → Repair or replace the faulted parts.
Yes ↓	

Step 8	Check the fuel temperature sensor(components).	
		1 Connect the fuel temperature sensor and the harness plug of engine control module. 2 Turn the ignition switch to ON position. 3 Measure the terminal voltage of the harness plug of engine control module. Measure circuit: the voltage between FB10 harness plug terminal 14 and ground. **Standard value: 3.1~3.3V(10 ℃)** **Standard value: 2.6~2.8V(20 ℃)** **Standard value: 2.1~2.3V(30 ℃)** 4 Check whether the result is normal?
		No → Replace the fuel temperature sensor.
Next ↓		
Step 9	Complete the check and confirm whether the system works normally nor not.	